

**AIMS**  
**Airbus Material Specification**

**Aluminium alloy (7475)**  
**Solution treated, controlled stretched**  
**and artificially aged (T7351)**  
**Plate**

**6,0 mm < a ≤ 100 mm**  
**Close tolerance flatness**

**Material Specification**

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## 1 Scope

This Material Specification defines the requirements of aluminium alloy 7475-T7351 plate  $6,0 \text{ mm} < a \leq 100 \text{ mm}$  for aerospace application.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

## 2 Application

No particular application

## 3 Normative references

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment or revision. For undated references, the latest issue of the publication referred to shall be applied.

|                |                                                                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIMS 03-02-000 | Airbus Material Specification – Technical specification, Aluminium and aluminium alloy wrought products – Plate                                                                |
| ABS 5052       | Aerospace series – Plates – Aluminium alloy 7475, close tolerance flatness<br>Thickness $6,0 \text{ mm} < a \leq 100 \text{ mm}$ , Dimensions                                  |
| EN 2032-2      | Aerospace series – Metallic materials –<br>Part 2: Coding of the metallurgical condition in delivery condition                                                                 |
| EN 4500-2      | Aerospace series – Metallic materials – Rules for drafting and presentation of material standards, Part 2: Specific rules for aluminium, aluminium alloys and magnesium alloys |
| EN 6072        | Aerospace series – Metallic materials – Test methods – Constant amplitude fatigue testing                                                                                      |

## 4 Technical requirements

### 4.1 General

Specification values are individual minimum except where stated.

### 4.2 Specific

See tables according to EN 4500-2

## 5 Designation

Refer to ABS 5052

## 6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS03-02-000.

|      |                                                      |         |                                                                                                                                                                                                             |      |     |      |     |      |      |      |        |       |      |
|------|------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|------|-----|------|------|------|--------|-------|------|
| 1    | Material designation                                 |         | Aluminium alloy 7475                                                                                                                                                                                        |      |     |      |     |      |      |      |        |       |      |
| 2    | Chemical Composition [%]                             | Element | Si                                                                                                                                                                                                          | Fe   | Cu  | Mn   | Mg  | Cr   | Zn   | Ti   | others |       | Al   |
|      |                                                      |         |                                                                                                                                                                                                             |      |     |      |     |      |      |      | each   | total |      |
|      |                                                      | min.    | -                                                                                                                                                                                                           | -    | 1,2 | -    | 1,9 | 0,18 | 5,2  | -    | -      | -     | Base |
| max. | 0,10                                                 | 0,12    | 1,9                                                                                                                                                                                                         | 0,06 | 2,6 | 0,25 | 6,2 | 0,06 | 0,05 | 0,15 |        |       |      |
| 3    | Method of melting                                    |         | —                                                                                                                                                                                                           |      |     |      |     |      |      |      |        |       |      |
| 4    | Form<br>Method of production<br>Limit dimension [mm] |         | Plate<br>Rolled<br>6,0 < a ≤ 100                                                                                                                                                                            |      |     |      |     |      |      |      |        |       |      |
| 5    | Technical specification                              |         | AIMS 03-02-000                                                                                                                                                                                              |      |     |      |     |      |      |      |        |       |      |
| 6    | 6.1 Delivery condition<br>and heat treatment         |         | T7351<br><br>Solution treated 465°C ≤ θ ≤ 475°C / WQ θ ≤ 40°C<br>+ controlled stretched (1,5 – 3 %)<br>+ two-step artificially aged:<br>105°C ≤ θ ≤ 125°C / 4h ≤ t ≤ 24h + 165°C ≤ θ ≤ 180°C / 5h ≤ t ≤ 20h |      |     |      |     |      |      |      |        |       |      |
|      | 6.2 Delivery condition code                          |         | U <sup>1)</sup>                                                                                                                                                                                             |      |     |      |     |      |      |      |        |       |      |
| 7    | Use condition and heat treatment                     |         | T7351 as delivered                                                                                                                                                                                          |      |     |      |     |      |      |      |        |       |      |
| 8    | Sample Test piece Heat treatment                     |         | T7351                                                                                                                                                                                                       |      |     |      |     |      |      |      |        |       |      |

|    |                                    |                   |                   |     |                  |    |             |    |                 |     |    |                 |     |    |                 |
|----|------------------------------------|-------------------|-------------------|-----|------------------|----|-------------|----|-----------------|-----|----|-----------------|-----|----|-----------------|
| 9  | Dimension concerned                |                   | a                 | mm  | 6 < a ≤ 25       |    | 25 < a ≤ 38 |    | 38 < a ≤ 51     |     |    | 51 < a ≤ 63     |     |    |                 |
| 10 | Thickness of cladding on each face |                   |                   | %   | —                |    |             |    |                 |     |    |                 |     |    |                 |
| 11 | Direction of test piece            |                   |                   |     | L                | LT | L           | LT | ST              | L   | LT | ST              | L   | LT | ST              |
| 12 | TENSILE                            | Temperature       | θ                 | °C  | Room temperature |    |             |    |                 |     |    |                 |     |    |                 |
| 13 |                                    | Proof stress      | R <sub>p0,2</sub> | MPa | 415              |    | 415         |    | 385             | 400 |    | 370             | 395 |    | 365             |
| 14 |                                    | Strength          | R <sub>m</sub>    | MPa | 490              |    | 490         |    | 460             | 480 |    | 455             | 475 |    | 450             |
| 15 |                                    | Elongation        | A                 | %   | 7                |    | 9           | 8  | 4 <sup>2)</sup> | 9   | 7  | 4 <sup>2)</sup> | 9   | 7  | 4 <sup>2)</sup> |
| 16 |                                    | Reduction of area | Z                 | %   | —                |    |             |    |                 |     |    |                 |     |    |                 |

|    |                     |                       |                |     |                   |
|----|---------------------|-----------------------|----------------|-----|-------------------|
| 17 | Hardness            |                       | HB             | —   | —                 |
| 18 | Shear strength      |                       | R <sub>c</sub> | MPa | —                 |
| 19 | Bending             |                       | k              | —   | —                 |
| 20 | Impact strength     |                       | k              | —   | —                 |
| 21 | CREEP               | Temperature           | θ              | °C  | —                 |
| 22 |                     | Time                  |                | h   | —                 |
| 23 |                     | Stress                | σ <sub>a</sub> | MPa | —                 |
| 24 |                     | Elongation            | a              | %   | —                 |
| 25 |                     | Rupture stress        | σ <sub>R</sub> | MPa | —                 |
| 26 |                     | Elongation at rupture | A              | %   | —                 |
| 27 | Notes (see Line 98) |                       |                |     | 1), 2), 3) and 4) |

|   |                                                      |         |                                                                                                                                                                                                                                                                                                                                                                                                            |      |     |      |     |      |     |      |        |      |      |
|---|------------------------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|------|-----|------|-----|------|--------|------|------|
| 1 | Material designation                                 |         | Aluminium alloy 7475                                                                                                                                                                                                                                                                                                                                                                                       |      |     |      |     |      |     |      |        |      |      |
| 2 | Chemical Composition [%]                             | Element | Si                                                                                                                                                                                                                                                                                                                                                                                                         | Fe   | Cu  | Mn   | Mg  | Cr   | Zn  | Ti   | others |      | Al   |
|   |                                                      | min.    | -                                                                                                                                                                                                                                                                                                                                                                                                          | -    | 1,2 | -    | 1,9 | 0,18 | 5,2 | -    | -      | -    |      |
|   |                                                      | max.    | 0,10                                                                                                                                                                                                                                                                                                                                                                                                       | 0,12 | 1,9 | 0,06 | 2,6 | 0,25 | 6,2 | 0,06 | 0,05   | 0,15 | Base |
| 3 | Method of melting                                    |         | -                                                                                                                                                                                                                                                                                                                                                                                                          |      |     |      |     |      |     |      |        |      |      |
| 4 | Form<br>Method of production<br>Limit dimension [mm] |         | Plate<br>Rolled<br>$6,0 < a \leq 100$                                                                                                                                                                                                                                                                                                                                                                      |      |     |      |     |      |     |      |        |      |      |
| 5 | Technical specification                              |         | AIMS 03-02-000                                                                                                                                                                                                                                                                                                                                                                                             |      |     |      |     |      |     |      |        |      |      |
| 6 | 6.1 Delivery condition<br>and heat treatment         |         | T7351<br><br>Solution treated $465^{\circ}\text{C} \leq \theta \leq 475^{\circ}\text{C}$ / WQ $\theta \leq 40^{\circ}\text{C}$<br>+ controlled stretched (1,5 – 3 %)<br>+ two-step artificially aged:<br>$105^{\circ}\text{C} \leq \theta \leq 125^{\circ}\text{C}$ / $4\text{h} \leq t \leq 24\text{h}$ + $165^{\circ}\text{C} \leq \theta \leq 180^{\circ}\text{C}$ / $5\text{h} \leq t \leq 20\text{h}$ |      |     |      |     |      |     |      |        |      |      |
|   | 6.2 Delivery condition code                          |         | U <sup>1)</sup>                                                                                                                                                                                                                                                                                                                                                                                            |      |     |      |     |      |     |      |        |      |      |
| 7 | Use condition and heat treatment                     |         | T7351 as delivered                                                                                                                                                                                                                                                                                                                                                                                         |      |     |      |     |      |     |      |        |      |      |
| 8 | Sample Test piece Heat treatment                     |         | T7351                                                                                                                                                                                                                                                                                                                                                                                                      |      |     |      |     |      |     |      |        |      |      |

|    |                                    |                   |                   |             |                  |    |                 |     |    |                 |     |    |                 |
|----|------------------------------------|-------------------|-------------------|-------------|------------------|----|-----------------|-----|----|-----------------|-----|----|-----------------|
| 9  | Dimension concerned                | a                 | mm                | 63 < a ≤ 76 |                  |    | 76 < a ≤ 89     |     |    | 89 < a ≤ 100    |     |    |                 |
| 10 | Thickness of cladding on each face |                   | %                 | –           |                  |    |                 |     |    |                 |     |    |                 |
| 11 | Direction of test piece            |                   |                   | L           | LT               | ST | L               | LT  | ST | L               | LT  | ST |                 |
| 12 | TENSILE                            | Temperature       | θ                 | °C          | Room temperature |    |                 |     |    |                 |     |    |                 |
| 13 |                                    | Proof stress      | R <sub>p0,2</sub> | MPa         | 385              |    | 365             | 365 |    | 350             | 360 |    | 345             |
| 14 |                                    | Strength          | R <sub>m</sub>    | MPa         | 470              |    | 450             | 450 |    | 440             | 440 |    | 435             |
| 15 |                                    | Elongation        | A                 | %           | 9                | 7  | 4 <sup>2)</sup> | 9   | 7  | 3 <sup>2)</sup> | 9   | 7  | 3 <sup>2)</sup> |
| 16 |                                    | Reduction of area | Z                 | %           | –                |    |                 |     |    |                 |     |    |                 |

|    |                     |                       |                |                   |   |
|----|---------------------|-----------------------|----------------|-------------------|---|
| 17 | Hardness            |                       | HB             | —                 | — |
| 18 | Shear strength      |                       | R <sub>c</sub> | MPa               | — |
| 19 | Bending             |                       | k              | —                 | — |
| 20 | Impact strength     |                       | k              | —                 | — |
| 21 | CREEP               | Temperature           | θ              | °C                | — |
| 22 |                     | Time                  |                | h                 | — |
| 23 |                     | Stress                | σ <sub>a</sub> | MPa               | — |
| 24 |                     | Elongation            | a              | %                 | — |
| 25 |                     | Rupture stress        | σ <sub>R</sub> | MPa               | — |
| 26 |                     | Elongation at rupture | A              | %                 | — |
| 27 | Notes (see Line 98) |                       |                | 1), 2), 3) and 4) |   |

|     |                                                  |   |                                                                                                                                             |                                                                     |     |                                                                                                                            |           |
|-----|--------------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------|-----------|
| 32  | Electrical conductivity                          | 7 | $\gamma$ [MS/m]                                                                                                                             | $R_{p0,2}$                                                          |     | Action                                                                                                                     |           |
|     |                                                  |   | $\gamma < 22,0$                                                                                                                             | Not acceptable                                                      |     |                                                                                                                            |           |
|     |                                                  |   | $22,0 \leq \gamma < 23,2$                                                                                                                   | $R_{p0,2 \text{ LT}} > R_{p0,2 \text{ LT min}} + 80 \text{ MPa}$    |     | – further artificial ageing or<br>– re-solution treatment or<br>– stress corrosion test or<br>– see footnote <sup>3)</sup> |           |
|     |                                                  |   |                                                                                                                                             | $R_{p0,2 \text{ LT}} \leq R_{p0,2 \text{ LT min}} + 80 \text{ MPa}$ |     | Acceptable                                                                                                                 |           |
|     |                                                  |   | $23,2 \leq \gamma \leq 26,0$                                                                                                                | Acceptable                                                          |     |                                                                                                                            |           |
|     |                                                  |   | $\gamma > 26,0$                                                                                                                             | Not acceptable                                                      |     |                                                                                                                            |           |
| 39  | Stress corrosion                                 | 3 | a > 25 mm                                                                                                                                   |                                                                     |     |                                                                                                                            |           |
|     |                                                  | 6 | ST direction, tension, constant load, length of exposure: 30 days,<br>stress level: 75% $R_{p0,2}$ specified in LT-direction                |                                                                     |     |                                                                                                                            |           |
|     |                                                  | 7 | Criterion: no failure                                                                                                                       |                                                                     |     |                                                                                                                            |           |
| 40  | Fracture toughness<br>$K_{IC}$ [MPa $\sqrt{m}$ ] | 7 | Dimension [mm]                                                                                                                              | L–T                                                                 | T–L | S–L                                                                                                                        |           |
|     |                                                  |   | $25 < a \leq 60$                                                                                                                            | 42                                                                  | 36  | –                                                                                                                          |           |
|     |                                                  |   | $60 < a \leq 100$                                                                                                                           | 44                                                                  |     | 27                                                                                                                         |           |
| 44  | External defects                                 | – | See AIMS 03-02-000                                                                                                                          |                                                                     |     |                                                                                                                            |           |
| 46  | Fatigue (open hole) <sup>4)</sup>                | 3 | T-type specimen in accordance with EN 6072                                                                                                  |                                                                     |     |                                                                                                                            |           |
|     |                                                  | 7 | The curve of the test results, calculated in accordance with EN 6072,<br>shall not lie below the reference curve defined by following data: |                                                                     |     |                                                                                                                            |           |
|     |                                                  |   | $\sigma_{\text{max nett}}$ [MPa]                                                                                                            | Number of cycles                                                    | R   | $K_t$                                                                                                                      | Direction |
|     |                                                  |   | 225                                                                                                                                         | 3 E04                                                               | 0,1 | 2,3                                                                                                                        | T–L       |
|     |                                                  |   | 140                                                                                                                                         | 1 E05                                                               |     |                                                                                                                            |           |
|     |                                                  |   | 120                                                                                                                                         | 2 E05                                                               |     |                                                                                                                            |           |
| 100 | 1 E06                                            |   |                                                                                                                                             |                                                                     |     |                                                                                                                            |           |
| 61  | Internal defects                                 | – | See AIMS 03-02-000                                                                                                                          |                                                                     |     |                                                                                                                            |           |
| 69  | Elastic modulus                                  | 3 | LT direction                                                                                                                                |                                                                     |     |                                                                                                                            |           |
|     |                                                  | 7 | The value must be within the following range<br>$70\,000 \leq E \leq 74\,000 \text{ MPa}$<br>$71\,000 \leq E_C \leq 75\,000 \text{ MPa}$    |                                                                     |     |                                                                                                                            |           |
| 70  | Compression Proof stress<br>$R_{0,2C}$ [MPa]     | 7 | Dimension [mm]                                                                                                                              | L                                                                   | LT  | ST                                                                                                                         |           |
|     |                                                  |   | $6 < a \leq 38$                                                                                                                             | 400                                                                 | 420 | 425                                                                                                                        |           |
|     |                                                  |   | $38 < a \leq 51$                                                                                                                            | 385                                                                 | 415 | 415                                                                                                                        |           |
|     |                                                  |   | $51 < a \leq 63$                                                                                                                            | 370                                                                 | 400 | 400                                                                                                                        |           |
|     |                                                  |   | $63 < a \leq 76$                                                                                                                            | 365                                                                 | 400 | 400                                                                                                                        |           |
|     |                                                  |   | $76 < a \leq 100$                                                                                                                           | 340                                                                 | 370 | 370                                                                                                                        |           |

|                       |                                                               |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                      |                               |                        |                  |           |
|-----------------------|---------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------------------|------------------|-----------|
| 71                    | Crack propagation <sup>4)</sup>                               | 3                   | CCT specimen, W = 160 mm, 2a <sub>i</sub> = 4,0 mm, 2a <sub>f</sub> = 53 mm, thickness = 5,0 mm, L–T and T–L direction                                                                                                                                                                                                                                                                                                               |                               |                        |                  |           |
|                       |                                                               | 7                   | The crack propagation rates shall not be greater than the following data:                                                                                                                                                                                                                                                                                                                                                            |                               |                        |                  |           |
|                       |                                                               |                     | ΔK [MPa√m]                                                                                                                                                                                                                                                                                                                                                                                                                           | da/dN [mm/cycle]              |                        | R                |           |
|                       |                                                               |                     | 5                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2,1 E-5                       |                        | 0,1              |           |
|                       |                                                               |                     | 10                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2,0 E-4                       |                        |                  |           |
|                       |                                                               |                     | 20                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1,5 E-3                       |                        |                  |           |
|                       |                                                               |                     | 30                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4,0 E-3                       |                        |                  |           |
| 82                    | Batch uniformity                                              | –                   | See AIMS 03-02-000                                                                                                                                                                                                                                                                                                                                                                                                                   |                               |                        |                  |           |
|                       |                                                               | 7                   | Electrical conductivity                                                                                                                                                                                                                                                                                                                                                                                                              | γ = 24,0 MS/m (typical value) |                        |                  |           |
|                       |                                                               |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                      | δ ≤ 1,0 MS/m per product      | Δ ≤ 1,5 MS/m per batch |                  |           |
|                       |                                                               |                     | or                                                                                                                                                                                                                                                                                                                                                                                                                                   |                               |                        |                  |           |
|                       |                                                               |                     | Hardness                                                                                                                                                                                                                                                                                                                                                                                                                             | 130 HB (typical value)        |                        |                  |           |
| δ ≤ 20 HB per product |                                                               | Δ ≤ 30 HB per batch |                                                                                                                                                                                                                                                                                                                                                                                                                                      |                               |                        |                  |           |
| 85                    | Bearing stress<br>F <sub>bry</sub> and F <sub>bru</sub> [MPa] | 7                   | Dimension [mm]                                                                                                                                                                                                                                                                                                                                                                                                                       | F <sub>bry</sub>              |                        | F <sub>bru</sub> |           |
|                       |                                                               |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                      | e/D = 1,5                     | e/D = 2,0              | e/D = 1,5        | e/D = 2,0 |
|                       |                                                               |                     | 6 < a ≤ 38                                                                                                                                                                                                                                                                                                                                                                                                                           | 560                           | 670                    | 705              | 910       |
|                       |                                                               |                     | 38 < a ≤ 51                                                                                                                                                                                                                                                                                                                                                                                                                          | 565                           | 670                    | 710              | 925       |
|                       |                                                               |                     | 51 < a ≤ 76                                                                                                                                                                                                                                                                                                                                                                                                                          | 560                           | 655                    | 695              | 905       |
|                       |                                                               |                     | 76 < a ≤ 100                                                                                                                                                                                                                                                                                                                                                                                                                         | 530                           | 615                    | 670              | 860       |
| 98                    | Notes                                                         | –                   | 1) See EN 2032-2<br>2) A4d<br>3) A sample from the relevant lot shall be solution heat treated for a minimum of 30 minutes at 470 ± 5°C and water quenched (50°C ≤ θ ≤ 70°C). The electrical conductivity shall be measured within 15 minutes after quenching. If the difference in the electrical conductivity of the sample compared with the relevant lot is ≥ 3,5 MS/m, then the lot is acceptable.<br>4) For qualification only |                               |                        |                  |           |
| 99                    | Typical use                                                   | –                   | –                                                                                                                                                                                                                                                                                                                                                                                                                                    |                               |                        |                  |           |
| 100                   | Qualification                                                 | –                   | Frequency of qualification testing is included in AIMS 03-02-000. Recommended thickness are 12 mm, 50 mm and 100 mm.                                                                                                                                                                                                                                                                                                                 |                               |                        |                  |           |

RECORD OF REVISIONS

| Issue        | Clause modified                                                                         | Description of modification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1<br>10/1997 | –                                                                                       | New standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2<br>02/2000 |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 3<br>03/2002 |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4<br>01/2010 | General<br>Paragraph 3<br>Line 6<br>Line 32<br>Line 46<br>Line 70<br>Line 82<br>Line 85 | Adapted to new template and editorial changes<br>Normative references updated<br>Stretch rate changed to 3%<br>Supplemented and wording changed to avoid misinterpretation<br>Test specification updated<br>Thickness ranges $76 < a \leq 89$ and $89 < a \leq 100$ merged (same requirements)<br>Reference to TS AIMS 03-02-000 added<br>Thickness ranges $51 < a \leq 63$ and $63 < a \leq 76$ merged (same requirements)<br>Thickness ranges $76 < a \leq 89$ and $89 < a \leq 100$ merged (same requirements) |
|              |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |